

26. VENTRAL MIDLINE REFERENCE

DURATION : 05:28

STUDY SHEET

COURSE SUMMARY

In this session, we explore the key role of the ventral midline in embryogenesis, where it serves as a vital reference that guides embryonic development. The notochord, as a central element, induces the formation of the neural tube and influences the ascent of the sacrum. We highlight the importance of this reference for metabolism, the structure of body systems, and its link to health issues such as osteoporosis. Embryonic coilings and the structuring of the midlines are discussed, as is their integration with the diaphragms and the formation of the body's various systems, thus emphasising the complexity and harmony of the developing embryo.

VENTRAL MIDLINE REFERENCE

The **ventral midline centre** is a fundamental element of the embryo. It is a key centre that allows the **genome** to execute its programme and organise the entire embryo.

Without this axis, there is no **vital reference**. In a therapeutic approach, the aim is to allow the body to regain this reference.

A growing child, even a baby, has a fundamental need for this reference, for example for **calcium metabolism**. The loss of this reference hinders the ability to restructure or integrate calcium, leading to degradation and, ultimately, to degenerative processes such as **osteoporosis**.

This axis is crucial for the **ascension of the sacrum** and its development. It generates a compressive force on the sacrum, linked to **sphenobasilar synchronisation (SBS)**, which influences balance, including at the level of the sternum.

The **notochord** is the basis of this ventral midline. It induces the formation of the dorsal midline, which will become the **neural tube**.

The neural tube is built from the notochord, which means that the notochord is the inducer of the formation of the **posterior nervous system**. This posterior nervous system will grow and converge towards the centre, forming the anterior midline.

All this movement integrates the **cranio-sacral axis**, the repercussions of which manifest on the anterior part, at the sternal level. The sternum, in this lineage, is symbolic of a **"stretching star"**.

THE EMBRYO AND ITS STRUCTURING

The embryo initially consists of an **endodermal layer**, a notochord with **lateral mesoblast**.

If one visualises the embryo by turning it over, from a lateral point of view, one observes a **neural plate** above the notochord. This neural plate will evolve into a **neural groove**.

The neural groove will reorganise the fluid structure of the mesoderm. The appearance of a **primitive lateral aorta** is a key step.

The internal mesoderm does not grow at the same rate as the neural plate, which causes **flexion of the embryo**. During this flexion, the neural tube forms to become the **posterior neural axis**.

COILING AND MIDLINE FORMATION

The restructuring of the internal fluid system leads to the appearance and organisation of the **lateral aortae**. The differential growth of these structures brakes the embryo forwards while pushing it from behind.

- This induces a **coiling of the embryo**, orchestrated by the aortae and the amniotic cavity.
- The **yolk sac** deposits on the embryonic pedicle, forming the **umbilical cord**.
- At the same time, the anterior part is drawn forwards, giving rise to the future sternum and heart.

This is how the three midlines are built:

- The **notochordal** (ventral midline).
- The **posterior midline** (that of the neural tube).
- The **anterior midline** (that of the positive tube).

This embryonic coiling, a complex process, is at the heart of embryonic organisation. All systems are organised around their **primordial fluid system**.

This precise synchronisation and these inductions delimit the embryo:

- The ventral midline induces the development of the dorsal midline.
- The dorsal midline induces the development of the medial midline.

INTEGRATION AND DEVELOPMENT OF SYSTEMS

These midlines are integrated into the different **diaphragms** (thoracic, cervical, etc.). There are at least seven diaphragms.

These structures are fundamental for the development of the following systems:

- **Urogenital**
- **Metabolic digestive**

- **Rhythmic**
- **Neurosensory** (or mineral, vegetal, animal)